

# ABDURAHAMAN SHEMSEDIN

Embedded Systems & Hardware Engineer | Edge AI | Co- Founder & CEO , Dynamo

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## PROFESSIONAL SUMMARY

Electrical and Computer Engineer (BSc 2019, MSc 2024) with 7+ years of hands-on experience designing, building and deploying commercial-grade hardware, industrial control systems and deep-learning models. Co-founded and scaled two Ethiopian ventures, Dynamo Gaming Machines and Dynamo Education, from concept to revenue. Skilled across FPGA, PLC, PIC, Arduino, ESP, STM32, Raspberry Pi and Jetson Nano, and in taking products from circuit board to metal and wood enclosure to installed machine. Winner of Jimma University Project of the Year and several national and continental awards. Motivated to build high-reliability aerospace and autonomous technology in Ethiopia that strengthens national self-reliance and positions the country as an African hub.

## CORE HARDWARE & EMBEDDED COMPETENCIES

|                           |                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| FPGA                      | Digital logic design, state machines and high-speed signal-processing pipelines in HDL.                               |
| PLC & Industrial Control  | Ladder-logic programming, VFD integration, relay logic and safety interlocks for motors and rides.                    |
| STM32 (ARM Cortex-M)      | Embedded C/C++ firmware: SPI, I2C, UART, timers and PWM for motor and actuator control.                               |
| PIC Microcontrollers      | Low-level C firmware, interrupt handling and robust sensor-interface design.                                          |
| Arduino / ESP32 / ESP8266 | Production controller firmware: coin-op logic, relay and LED driving, USB HID devices, Wi-Fi/Bluetooth IoT telemetry. |
| Raspberry Pi              | Embedded Linux setup and Python/C++ control software for custom devices.                                              |
| NVIDIA Jetson Nano        | Edge AI deployment and OpenCV computer-vision pipelines for real-time detection and tracking.                         |

## PROFESSIONAL & COMMERCIAL EXPERIENCE

**Dynamo Centre for Technology** | *Co-Founder & Chief Executive Officer* 2019 – Present

- Dynamo Gaming Machines (15M+ ETB capital):** Engineered and manufactured commercial-grade arcade equipment, motion-simulators, driving rigs, and interactive amusement machines.
- Designed custom microcontroller-based input/output controller boards (RP2040, Arduino) to interface arcade peripherals, coin acceptors, potentiometers, switches, and solid-state relays.
- Programmed custom low-latency C++ firmware and configured specialized emulation wrappers (TeknoParrot, Lindbergh Loader, Model 2) to run hardware-bound software on custom x86 architecture.
- Oversee full-stack hardware fabrication, including metal/wood structural workshops, power supply wiring, motor rewiring, and VFD installation.
- Secured commercial installation and maintenance contracts with major entertainment centers (Yellow Building Fun Zone, Day Line Fun Zone, MELA Playground Equipment, Science Cafe, Aba Fun Zone).
- Dynamo Education (10M+ ETB):** Afan Oromo academic platform for Grades 7–12 on a Moodle-based LMS and mobile app (200+ daily users), with 3,000+ hours of video lessons, a professional recording studio, live tutoring and exam preparation.
- Trained 20,000+ youth and children in STEM and programming across Jimma through Dynamo’s short-term technical courses.
- Early in Dynamo, supervised and consulted on 80+ MSc and BSc hardware projects (see below).

## SELECTED ENGINEERING & RESEARCH PROJECTS

**Interpretable Deep Learning for Amharic Hadith Authenticity** | *MSc Thesis* 2021 – 2024

- Built and annotated a 16,654-text dataset across 12 authenticity grades; CNN-BiLSTM model reached 89% accuracy and outperformed baselines.
- Applied SHAP explainability to expose the features driving predictions. Final stage of publication in Nature journal with Dr. Kinde Anlay.

**Smart Solar-Powered Autonomous Tractor for Coffee Farms** | *BSc Final Project* 2019

- Institute Project of the Year at Jimma University Institute of Technology (earned a funded MSc place); Winner, Green Innovation & Agri-Tech Slam 2019.

Technology Transfer Projects | Jimma University Institute of Technology2016 – 2017

- **LED-matrix smart teaching device for kindergarten (2016):** Amharic Unicode encoding, speech-recognition hardware tuned on Amharic and Afan Oromo with many children’s voices.
- **Amharic Braille keyboard (2017):** hardware-to-OS integration and a 300+ word voice navigation dataset.
- **Low-cost digital advertising board (2016):** Amharic character decoding and animations for a custom LED matrix.

Engineering Consultancy for 80+ Student Projects | Dynamo2019 – 2023

- Award-winning builds include a digital spirometer to Ethiopian standards, Ultrasonic based dialysis machine, a computer-vision pesticide robot, an Amharic voice-synthesis book reader for the blind, Amharic sign-language gloves, a urine-based cervical cancer detector, an eye-tracking driver sleep detector, a VR simulator lab, an automated wheelchair and a driving simulator.

EDUCATION

MSc, Computer Engineering | Jimma University Institute of TechnologyGraduated 2024

BSc, Electrical and Computer Engineering | Jimma University Institute of TechnologyGraduated 2019

AWARDS & RECOGNITION

- Top 10 Finalist, Innovating Education in Africa (IEA), African Union (2025)
- MasterCard Foundation EdTech Fellowship (2025)
- UNICO Best Emerging Startup and Promising Technology (2023)
- TotalEnergies Startupper of the Year (2022)
- Green Innovation and Agri-Tech Slam, Winner (2019); Jimma University Institute of Technology Project of the Year (2019)

INTERNATIONAL EXPOSURE

- African Startup Conference, Algiers, Algeria (2025), facilitated by the African Union
- Learning Science and AI for EdTech conference, Prague, Czech Republic, hosted by Czech UNDP
- Agriculture Mechanisation Festival, Abidjan and Yamoussoukro, Côte d’Ivoire (2022), hosted by GIZ
- Experience-sharing visits to Saudi Arabia, Turkey and Egypt

REFERENCES

Kifle Godana Heyi (B.Sc., M.Sc.)

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